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REVIEW

Evidence-Based Management of Patellar Tendinopathy in Athletes: A Narrative Review

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Abstract

Purpose of Review To synthesize the available literature for evidence-based management of patellar tendinopathy (PT) among athletes and to provide a conceptual framework to help guide clinical decision-making when managing in-season and off-season athletes with PT.

Recent Findings Physical therapy remains the first line treatment for PT. Several studies have reported no improvement or inferior outcomes in patients treated with corticosteroid injection. Preliminary studies show promising outcomes for athletes treated with tendon scraping and percutaneous ultrasonic tenotomy. Recent studies report high rates of return to sport with arthroscopic and open surgical treatments in refractory cases.

Summary Despite PT being a common condition affecting athletes, the optimal management remains undetermined. The location and severity of patellar tendon pathology, individual characteristics of the affected athlete, timeline of the athletic season, recovery time post-intervention, treatment efficacy, associated cost, and potential treatment side effects should be considered when determining management. Multiple effective and safe non-operative treatment options are available to manage this condition with surgery available for refractory cases. The treatment framework described here is based on a comprehensive review of the available evidence.

Keywords Patellar tendinopathy · Athletes · Platelet rich plasma · Tendon scraping · Rehabilitation · Percutaneous ultrasonic tenotomy

Introduction

Patellar tendinopathy (PT), or "jumper's knee," is an overuse injury affecting up to 18.3% of athletes compared to 0.1% of the general population [1]. It results from repetitive mechanical loading of the patellar tendon, leading to degenerative changes and impaired tendon healing response, which cause pain and dysfunction [2]. Emerging evidence also suggests a genetic predisposition [3]. Clinically, PT presents with gradual onset pain at the inferior patellar pole that is exacerbated with movements that load the knee extensor mechanism, often during periods of increased sport-related training. Diagnosis is supported by ultrasound (US) or magnetic resonance imaging (MRI) findings such as tendon thickening, disrupted tendon structure, enthesopathic changes, and neovascularity [4].

Treatment options for athletes with PT include physical therapy, bracing/taping, medications, injections, percutaneous interventions, extracorporeal shockwave therapy (ESWT), and surgery. However, no consensus exists regarding the optimal sequence or timing of these interventions. Limited comparative data and multiple variables – chronicity of symptoms, severity, prior treatments, sport-specific demands, cost, and season timing – complicate clinical decision-making, particularly in refractory cases.

This review summarizes current non-operative and operative treatments for athletes with PT and offers a clinical framework to guide management, with attention to in-season versus off-season considerations.

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Methods

A comprehensive search was conducted of MEDLINE/PubMed and EMBASE electronic databases for studies published from 2015 to May 2025. Key search terms included patellar tendinopathy, patellar tendinosis, patellar tendinitis, jumper's knee, athletes, physical therapy, exercise, extracorporeal shockwave therapy, knee brace, patellar strap, athletic tape, corticosteroid, platelet rich plasma, bone marrow aspirate concentrate, hyaluronic acid, prolotherapy, sclerotherapy, tendon scraping, tenotomy, debridement, ultrasound, surgery, arthroscopic, and open debridement.

Inclusion and exclusion criteria are outlined in Table 1. Articles were selected based on relevance to PT in athletes emphasizing original research and high-quality reviews.

All retrieved abstracts ($n=241$) were screened for this review, and articles not meeting the inclusion criteria were excluded. Reference lists from the selected studies and review articles were further assessed for eligibility.

Table 1 Inclusion and exclusion criteria

	Inclusion Criteria	Exclusion Criteria
Interventions	- Surgical debridement - Surgical resection - Injectables - Bracing/straps - Physical therapy - Glyceryl Trinitrate - Ultrasonic debridement - Percutaneous Needle Tenotomy - Extracorporeal shockwave therapy	
Population	- Human studies - All age of participants	- Animal studies
Condition	- Patellar tendinopathy (tendinitis, tendinosis) - Partial thickness patellar tendon tear	- Patellar tendon rupture - Full thickness patellar tendon tear - Related to any pathology other than patellar tendon (i.e. Sinding-Larsen-Johansson syndrome, Osgood-Schlatter disease, etc.)
Study Design	- Randomized control trials - Systematic reviews - Umbrella reviews - Scoping reviews - Cohort studies - Case series - Case studies - Study is in English	- Studies not in English

Results

Non-Operative Treatment Options

Physical Therapy

Physical therapy is often a first-line treatment option for athletes with PT. Physical therapy can include static and dynamic stretching, isometric, concentric and eccentric-focused strength training, progressive tendon loading, and heavy slow resistance training. Victorian Institute of Sport Assessment – Patellar Tendon (VISA-P) scores are patient-reported outcome measures to assess symptom severity, functional limitation, and sport participation in those with PT [5]. A higher score (range 0–100) indicates less symptom severity, higher function, and ability to participate in sport.

Stretching

Stretching is not usually indicated in isolation for the management of PT, but some studies have evaluated its effect in combination with other treatments. Dimitrios et al. found superior improvements in VISA-P scores when static stretching was added to an eccentric strengthening protocol [6]. Gatz et al. found successful clinical outcomes were achieved by patients performing quadriceps, hip flexor, hamstring, and calf stretches in addition to topical polidocanol [7]. Typical treatment parameters involve stretching 1 to 3 times per day, holding each position for 30 s. While there is a paucity of literature supporting static stretching in isolation for PT, it may be a useful adjunct to other strengthening protocols. Caution should be exercised when prescribing stretching exercises to avoid excessive tendon compression against adjacent bone (i.e., patellar enthesophyte), which may lead to further tendon pathology [8].

Eccentric Exercise

Eccentric-focused training has long been the foundation of PT management, and several studies have reported improved pain and function. Multiple systematic reviews and network meta-analyses report eccentric training as being among the most effective treatments for PT, particularly when evaluating improvements in VISA-P and Visual Analog Scale (VAS) pain scores [9–11]. Compared to concentric training and eccentric step-down training, the eccentric slant board exercise protocol results in significantly better long-term pain improvement [12, 13]. The decline squat eccentric program has also been shown to have positive effects on

patellar tendon health, including reduced tendon stiffness and increased tendon strain [14]. Although there is variability among eccentric exercise protocols in the literature, most programs involve single or double leg eccentric slant board squats prescribed for 3 sets of 15 repetitions, often 2 times per day, with a goal of maintaining tendon pain levels less than 5/10 on the VAS pain score during exercise [11, 12, 14, 15]. It remains important for the clinician to monitor patient response to this type of program and adjust the exercise dosage as necessary.

Isometric Exercise

Isometric knee extensor exercises, which involve the quadriceps myotendinous unit producing force without associated osteokinematic knee joint motion, are often prescribed for athletes with PT. Rio et al. and van Ark et al. describe improvements in VAS pain scores and VISA-P scores following 4 weeks of isometric exercise [16, 17]. Pearson et al. reported improvement in pain levels for athletes treated with short-duration isometric holds (10 s) and athletes treated with long-duration isometric holds (40 s) for 4 weeks, though the study was underpowered to detect differences between the two groups [18]. Holden et al. also found acute analgesic effects following isometric exercise, though these effects did not persist past 45 min [19]. Several studies show that isometric exercises do not outperform isotonic exercises in the management of chronic tendinopathy [10, 20, 21]. A common limitation of previously studied isometric interventions has been a short follow-up period of 4 weeks [16–18]. While isometrics may be helpful for initiating tendon loading in highly symptomatic patients and in-season athletes due to the potential analgesic effects, this should not be the sole exercise selection for athletes with PT in the medium- to long-term.

Progressive Tendon Loading Exercise

More recently, progressive tendon loading exercise (PTLE) and heavy slow resistance (HSR) have been evaluated as alternative protocols to eccentric training for athletes with PT. The use of PTLE for PT was initially described by Malliaris et al. [22]. This approach stages different exercise types for tendinopathy management beginning with isometrics, progressing to isotonic exercise, and finally ending with energy storage and release tasks (i.e., plyometrics). Breda et al. found that this treatment led to significant improvements in VISA-P and VAS pain scores, with significantly better VISA-P scores in the PTLE group compared to the eccentric training group at 24 weeks post-intervention [23]. The PTLE exercises were also significantly less painful compared to the eccentric group exercises, highlighting a potential role for using PTLE when treating in-season athletes. It is important to note that many athletes in both groups reported clinically significant change in symptoms much earlier than 24 weeks, with 49% of athletes in the PTLE group and 55% of athletes in the eccentric training group reporting improvement of 13 points or better on the VISA-P at 12 weeks post-intervention [23]. A recent 5-year follow-up of this study demonstrated 76% of participants being defined as recovered, while 24% did not feel recovered [24]. Despite no clear prognostic factors identified for self-reported recovery, this study highlights some cases may be refractory and require additional treatment.

Heavy Slow Resistance Exercise

HSR protocols typically consist of weighted squats and leg press completed with a 3-second eccentric phase and a 3-second concentric phase (i.e. 6 s/repetition) [25]. Kongsgaard et al. found that patients treated with HSR reported improved VISA-P and VAS pain scores within 12 weeks and maintained up to 6 months. Compared to eccentric training, the HSR group had significantly higher satisfaction scores [25]. Ruffino et al. compared HSR to flywheel training and found significant improvement in VISA-P scores in both groups with no significant difference between the two training modalities [26]. HSR training can be considered for athletes with PT, with available evidence suggesting comparable outcomes compared to eccentric training and one study reporting increased patient satisfaction [25].

Inertial Flywheel Training

Inertial flywheel training utilizes the momentum of a spinning wheel, created during the concentric phase of a repetition, to increase inertial resistance when braking during the eccentric phase [27]. As previously described, Ruffino et al. found comparable improvements in VISA-P scores when comparing flywheel training to HSR [26]. Romero-Rodriguez et al. also reported significant improvements in VISA-P and VAS scores in a case series of 10 male athletes treated with 6 weeks of inertial flywheel training [27]. Though the evidence is limited for inertial flywheel training as an isolated treatment, this can be considered in conjunction with other physical therapy training modalities.

Concentric Exercise

Only one study to date has assessed the outcomes of concentric-only training for patients with PT [12]. In the concentric-only group, all 9 patients were dissatisfied with the

treatment and reported no change in VAS pain or VISA-P scores. Therefore, concentric-only training does not seem to have a role in the treatment of athletes with PT.

Bracing and Taping

Athletes with PT often use patellar straps and sports tape with the goal of improving pain during sport. Recent studies have evaluated the effect of these modalities on pain, biomechanics, and performance [28–32].

In a randomized controlled trial (RCT), De Vries et al. reported improved VAS pain scores two hours after sports participation in athletes treated with a patellar strap, sports tape, and placebo tape (did not conform with kinesiotaping guidelines). There was also a control arm with no intervention provided [28]. During this study, the sports tape group was treated with non-elastic tape applied perpendicularly over the patellar tendon and wrapped circumferentially around the proximal tibia under slight tension; the placebo tape group received a short piece of elastic tape applied perpendicularly over the patellar tendon with minimal tension in a non-functional manner [28]. Rosen et al. found that patellar strap use during single-limb landing was associated with reduced pain, a more neutral landing biomechanical alignment, and decreased landing forces [29]. When evaluating patellar strap use in young male basketball and volleyball athletes, Dar et al. found similar improvements in pain without a negative effect on jump performance parameters [30]. Tamura et al. found a significant reduction in pain with sports taping in college-age females, but sports tape use in this study was also associated with a decreased maximum jump height [31].

Current evidence suggests patellar straps can be safely recommended to athletes with PT to reduce pain with sport-related activities and potentially improve landing biomechanics. While evidence is limited regarding the effect of these modalities on performance, athletes using sports tape should be counseled that it may contribute to a slight decrease in maximum vertical jump height [31].

Topical Glyceryl Trinitrate

Topical glyceryl trinitrate (GTN), also known as nitroglycerin, has been studied as a treatment for a variety of tendinopathies [33–36]. Topical nitroglycerin is hypothesized to contribute to tendon healing by delivery of nitric oxide locally, which may enhance local blood flow, collagen synthesis, and fibroblast activity to improve tendon structure and function [34, 36, 37]. This medication is delivered though a topical patch applied to the affected area once daily. Typical doses of the topical patches reported in the literature range from 1.25 to 5 mg per day with a duration of 3 days to 6 months [33, 34].

In a systematic review of GTN use for multiple chronic tendinopathies, Challoumas et al. reported improvement in short-term (< 8 weeks) and intermediate-term (< 6 months) pain and increased patient satisfaction when compared with placebo [34]. However, the only RCT included in this systematic review that directly evaluated GTN patch for PT, completed by Steunebrink et al., found no difference between eccentric exercise with a GTN patch versus eccentric exercise with a placebo patch [33]. Van Rijn et al. performed a secondary analysis of the results reported by Steunebrink et al. and found that eccentric training alone was more likely to achieve clinical improvement than eccentric training combined with topical GTN [38].

In two subsequent systematic reviews with network meta-analyses Challoumas et al. reported that topical GTN combined with eccentric exercise was among the interventions with the highest probability of being effective for the treatment of PT in spite of the fact that their network meta-analyses included the study by Steunebrink et al. which found no improvement with GTN [9, 39].

While the mechanistic rationale for topical nitroglycerin is plausible, significant improvement in patient-reported outcomes has not been found. However, it may be considered as an adjunct non-invasive option for athletes who fail to respond with physical therapy alone. Consideration of medical comorbidities and potential side effects, such as headaches, lightheadedness, and skin rashes, should be considered, and athletes should be counseled to monitor for these symptoms [21, 33, 34].

Focused Shockwave and Radial Pressure Wave

Extracorporeal shockwave therapy (ESWT), historically referring to focused shockwave (FSW) and radial pressure (RPW), has been utilized for a variety of musculoskeletal conditions. A recent Delphi consensus statement highlighted the need to more appropriately reserve ESWT for FSW and independently specify RPW [40]. The proposed mechanism of action involves the application of high-energy acoustic waves to generate mechanical and cellular responses aimed at promoting cellular migration and proliferation, increased vascularity, and altering pain signaling pathways at the site of damaged tissue [41, 42]. FSW and RPW each produces a characteristic wave form with RPW having the greatest energy and pressure near the applicator source, while FSW produces a shockwave that propagates deeper within tissue [41, 42].

Several recent systematic reviews have evaluated the efficacy of ESWT for the treatment of PT with mixed outcomes. Some meta-analyses have shown a modest improvement in pain reported in groups treated with ESWT [43–45]

while others have shown no difference compared to placebo or control treatments [9, 46, 47].

There are many variables to account for when evaluating and comparing efficacy of shockwave treatments for musculoskeletal conditions including the device used, FSW or RPW, frequency and number of sessions, energy level, number of impulses, and head type used. The reported treatment protocols vary widely among the RCTs included in these systematic reviews. For example, Cheng et al. [48] report significant improvement in pain and function after 16 weekly sessions of RPW, while Thijs et al. [49] report no significant differences in outcomes between three sessions of FSW or sham treatment combined with eccentric exercise. The heterogeneity in protocols utilized for treatment of PT make it challenging to draw definitive conclusions regarding efficacy or optimal treatment protocols.

Several studies evaluating concurrent treatment with FSW or RPW and eccentric exercise found no difference when compared to eccentric exercise combined with sham [14, 38, 49]. Similarly, Zwerver et al. found no benefit with 3 sessions of FSW at 1-week intervals compared to placebo for in-season jumping athletes [50]. However, an RCT by Jhan et al. found that athletes who received combined treatment of PRP with one session of FSW administered one week later reported faster reduction of pain when compared to athletes receiving PRP with sham FSW treatment [51].

The involvement of changes to Hoffa's fat pad may also influence the effectiveness of this treatment. Williams et al. described a retrospective study in which 17 of 20 patients with tendinopathic changes isolated to the patellar tendon on MRI achieved good or excellent improvement in symptoms with RPW while all patients with MRI changes (e.g. fat pad thickening or T2 signal intensity suggestive of edema) extending to Hoffa's fat pad failed treatment with RPW and ultimately required arthroscopic debridement [52]. While involvement of Hoffa's fat pad may be a risk factor for less favorable results with RPW, the RPW protocol was not outlined and therefore is hard to reproduce or assess for adequate dosing. Current studies demonstrate mixed results for the use of FSW and RPW in isolation or in combination with other treatment modalities, for athletes with PT. Optimal protocols for RPW and FSW have yet to be established. Regardless, FSW and RPW remain safe treatment options that can be considered in combination with physical therapy. Furthermore, RPW and FSW are favorable options to consider for in-season athletes due to minimal post-procedure activity restrictions while promoting a healing response. This is in comparison to alternative options, such as PRP, which may require a short duration away from sport post-procedure. While no high-level studies exist to support the use of combined RPW and FSW, clinical experience suggests that combining these two treatments may take advantage of unique cellular responses that may lead to improved outcomes. However, other factors, including the cost, need for multiple sessions, and imaging characteristics (e.g. high-grade partial thickness tear), should be considered.

Corticosteroid Injection

Corticosteroid injections are not currently recommended for the management of PT. A recent network meta-analysis by Wang et al. showed inferior VISA-P and VAS pain scores 6 months after corticosteroid injection when compared to non-injection treatment options [53]. Several additional RCTs and systematic reviews have found no long-term benefit following treatment with corticosteroid injection [9, 25, 54, 55]. Furthermore, peritendinous corticosteroid injection can potentially increase the risk of tendon rupture, particularly in athletes whose sport activities require repetitive high impact loading of the patellar tendon. Chen et al. described seven weightlifters who suffered patellar tendon ruptures after multiple peritendinous corticosteroid injections [56]. Given the potential increased risk of tendon rupture and evidence of poor long-term treatment outcomes, corticosteroid injections should not routinely be used for the treatment of PT in athletes.

Platelet-Rich Plasma

Platelet-rich plasma (PRP) is an autologous product of whole blood containing supraphysiologic levels of platelets that is thought to promote healing through local delivery of growth factors, cytokines, and chemokines [57, 58]. PRP has shown promise in improving pain and function in non-comparative studies [59–61]. However, higher level studies have produced mixed results when comparing PRP to control groups [62–64]. These mixed results may be due to a variety of factors, including platelet dose/concentration factor, leukocyte content, number and timing of injections, extent of needle tenotomy, and post-procedure rehabilitation.

PRP has been compared to other treatment modalities with varying outcomes. In an RCT of athletic individuals, Vetrano et al. compared 2 injections of PRP to 3 sessions of FSW (2400 impulses at 0.17–0.25 mJ/mm2 per session) at 48- to 72-hour intervals [65]. They found that while both treatments had significant improvement in symptoms, PRP had significantly better improvement at 6 months. A recent study by Jahn et al. found that combining PRP with FSW may improve short-term outcomes [51].

Studies have also evaluated the effect of PRP with concurrent percutaneous needle tenotomy (PNT). Dragoo et al. observed significant improvements at 12 weeks with the injection of leukocyte-rich PRP (LR-PRP) in addition to 10 needle fenestrations when compared to the control group

who received 10 needle fenestrations alone; however, the treatment group was also noted to be significantly younger than the control group [66]. In contrast, Herrero et al. found no added benefit with the addition of leukocyte-poor PRP (LP-PRP) to PNT [67].

Comparisons between PRP and high-volume injections (HVI) with local anesthetic and normal saline have demonstrated improvement in pain and function with both interventions [63], and there is some evidence of a synergistic effect when used in combination [68]. The optimal number of PRP injections remains unclear as prospective studies have produced mixed results [60, 69], but a meta-analysis by Andriolo et al. suggested multiple injections may be superior for long-term outcomes [45]. The optimal platelet dose and concentration factor are uncertain. These factors may influence the number of injections needed for therapeutic effect as higher dose PRP may only require a single injection [70].

Wide variation exists in PRP preparation across studies, including differences in injectate volume (3.5–6.5 mL), platelet concentration factor (1.6-10x baseline), platelet dose, leukocyte content, and the addition of PNT [59, 60, 62, 65–69, 71, 72]. It is worth noting that the majority of these studies did not individually measure platelet concentrations of the samples but rather estimated based on volume or machine characteristics, which may lead to false assumptions about the content of the PRP injectate [73]. Both LR-PRP and LP-PRP have demonstrated clinical benefit [66, 67], and no significant differences in outcomes were found between the two in a comparative study [72].

Few studies have specifically focused on athletic return-to-sport outcomes. Across several studies, approximately 80% of active patients and athletes report returning to pre-injury activity levels following PRP treatment [59, 61, 69]. However, Kaux et al. found this number to be closer to 55% at one year [60]. Zayni et al. reported 86% of athletes were able to return to pre-symptom sporting level at 6.7 weeks, which aligns with findings from a review by Liddle et al. [61, 69].

Current evidence suggests that PRP can be safely utilized to treat athletes with PT. However, when considering the out-of-pocket cost, time to pain and functional improvement, potential for more than one injection, and variable outcomes reported in the literature, PRP should be reserved for athletes with PT refractory to physical therapy. Given the recovery time involved, often involving several months, PRP may not be practical for the in-season athlete. While there are no comparative data regarding peritendinous versus intratendinous placement of PRP for athletes in-season, peritendinous injection (perhaps with HVI) could be considered to minimize postprocedure pain and recovery. In summary, PRP is a safe option for refractory PT treatment, but many uncertainties remain regarding optimizing its efficacy, and athletes should be counseled accordingly.

Hyaluronic Acid

Hyaluronic acid (HA) is a naturally occurring glycosaminoglycan found in synovial fluid and is commonly used in the treatment of knee osteoarthritis. While less extensively studied for tendinopathies, preclinical and small clinical trials suggest that it may offer benefits through inhibition of inflammatory pathways, increased cell viability and proliferation, improved tendon viscoelasticity, and optimizing of the collagen makeup of a tendon [74, 75].

Several case series have reported symptomatic improvement following peritendinous HA injections. Kumai et al. observed improvement at 1 week post injection [76], while Muneta et al. found that palpation-guided injections, sometimes requiring up to 11 treatments, led to excellent outcomes in 54% of athletes [77]. In addition to symptomatic improvement, HA injections may improve sonographic findings of PT [78]. A comparative study by Kaux et al. found similar improvement in symptoms between one intratendinous LP-PRP and two HA peritendinous injections [79].

Current evidence suggests that HA can be safely recommended to athletes for treatment of PT, though larger comparative studies are needed to assess its efficacy compared to other treatments. Additionally, HA is not FDA approved for this indication, and therefore, may not be covered by insurance.

High-Volume Injection

High-volume injection (HVI) refers to injecting a large volume (40–50 mL) of normal saline and local anesthetic (with or without corticosteroid) between the patellar tendon and Hoffa's fat pad to disrupt the pathologic neovascularity of the tendon and fat pad often seen in chronic PT [80, 81].

In a combined retrospective and prospective case series, Morton et al. found that HVI (10 mL 0.5% bupivacaine, 25 mg hydrocortisone, and 30 mL normal saline) followed by eccentric loading program led to improvement in mean VISA-P scores, though only 3 of the 8 prospective athletes in their study were able to return to sport within 12 weeks [82].

Maffulli et al. described 44 patients treated with HVI (10 mL 0.5% bupivacaine, 62,500 IU aprotinin, 40 mL normal saline) followed by an eccentric loading program and repeat injection if symptoms persisted at 2 weeks. They found clinically significant improvements in VISA-P and VAS at 15 months, with 23 of the 32 physically active patients returning to sport [80]. Similarly, a retrospective case series by Nielsen et al. showed an improvement in mean VISA-P score 15 months after a single HVI (10 mL 0.5% Marcaine, 0.5 mL triamcinolone, and 40 mL normal saline) and HSR training protocol, though only 32% of participants reported clinically significant improvement in VISA-P [83].

As previously described, Abate et al. reported a possible synergistic effect of combining HVI (10 mL 2% mepivacaine, 30 mL normal saline) with PRP, noting improved pain and VISA-P scores at 6 months in patients receiving two rounds of HVI + PRP when compared to patients receiving two rounds of either PRP or HVI alone [68].

Despite the encouraging results of prior studies, a recent double-blind RCT by Barker-Davies et al. evaluated HVI (10 mL 0.5% bupivacaine, 30 mL normal saline) for the treatment of patellar and Achilles tendinopathies and found no difference between HVI (both with and without 25 mg hydrocortisone) when compared to placebo injections [81].

Available evidence suggests HVI is a safe treatment option, though further controlled/comparative studies are warranted to better characterize the efficacy of this procedure. Most studies to date have combined HVI with other potential therapeutic agents (e.g. PRP), making it difficult to discern its therapeutic effect alone. Additionally, the optimal volume and/or role for repeat HVI remains unclear. While some athletes may have neovascularization with accompanying neonerves involving Hoffa's fat pad or the deep fibers of the patellar tendon, this phenotype does not represent all cases of PT. Nonetheless, this treatment option is safe and can be considered for in-season athletes as the lack of tendon trauma results in minimal recovery.

Prolotherapy

Prolotherapy (PrT) refers to the injection of an irritant, typically a hypertonic dextrose solution ranging in concentrations from 12.5% to 25%, at the site of pathology with the goal of stimulating a healing response and reducing pain [84, 85]. PrT has previously been used for the treatment of a variety of common sports-related tendinopathies, though few studies have evaluated its use for the treatment of PT [85].

In a prospective pilot study by Ryan et al., patients reported statistically significant improvements in VAS pain scores at rest, with activities of daily living, and with sport activities after receiving a median of four injections targeting intratendinous areas of abnormal hypoechogenicity or patellar tendon thickening [86].

In a recent systematic review evaluating the efficacy of PrT for several common sport-related tendinopathies, no significant adverse events were described in any of the included RCTs [85].

While limited, the available evidence suggests that PrT is safe for the treatment of PT in athletes, though further studies are needed to assess its efficacy compared to other treatment options. Furthermore, the need for multiple injections, potential for postprocedural pain flare, out-of-pocket cost, and intratendinous site of injection may limit the use of PrT for the in-season athlete [84].

Bone Marrow Aspirate Concentrate

Bone marrow aspirate concentrate (BMAC) is an autologous cellular therapy that is thought to release growth factors, initiate a paracrine effect from mesenchymal stromal cells, and lead to modulation of inflammation [87]. Preclinical studies have demonstrated the ability of BMAC to augment tendon repair, though human studies are lacking [88]. Pascual-Garrido et al. demonstrated potentially long-lasting effect of treatment for PT, with 7/8 subjects reporting being satisfied with treatment at 5 years [89]. In a double-blind, randomized study comparing LP-PRP to culture-expanded bone marrow mesenchymal stromal cells (BM-MSCs) expanded from bone marrow aspirate, Rodas et al. [90] observed improvement in symptoms in both groups at 6 months with no difference between the two groups. The BM-MSC group also demonstrated improvement in tendon structure on MRI compared to LP-PRP [90]. Despite promising results, culture-expanded BM-MSCs are not available to use in the United States outside of approved clinical trials and therefore are not a readily available option.

There is a paucity of literature utilizing BMAC for PT in athletes. Considering the typically high out-of-pocket cost, invasiveness, associated downtime, and need for additional studies on comparative outcomes, the routine use of BMAC for managing PT in athletes who have not already failed other conservative measures is not recommended.

Tendon Scraping

Ultrasound-guided percutaneous tendon scraping is a minimally invasive procedure that targets disruption of accompanying neoverves at sites of neovascularization by mechanical disruption. A blunt tool such as a meniscotome is utilized to separate the patellar tendon from the adjacent Hoffa's fat pad [91, 92].

Hall and Rajasekaran first described a case of a competitive high school athlete who was able to return to full, pain-free participation four weeks after the procedure [91]. In a subsequent retrospective case series by Baria et al. [92], five of six athletes available for follow-up reported clinically significant improvements in patient reported outcome measures at final evaluation. All competitive athletes in their study successfully returned to sport at one month, and no major adverse events were reported [92].

Despite the promising results reported thus far in the literature, no high-level or long-term studies are available for this treatment. As tendon scraping does not penetrate the tendon and involves minimal recovery, it can be considered for in-season athletes. However, longer-term and comparative studies are needed to understand the durability of this treatment and how it compares with other interventions.

Percutaneous Needle Tenotomy

Percutaneous needle tenotomy (PNT) involves repeated needle fenestration of tendinopathic tissue with a goal of inducing a local inflammatory response to promote tissue healing [93].

Housner et al. [94] described a retrospective study of 47 cases of PT in 32 patients treated with 20 to 30 ultrasound-guided fenestrations with a 22-gauge needle. At an average follow-up of 45 months, 72% of cases returned to sport with no pain or mild pain, and 81% of patients reported excellent or good satisfaction scores. One athlete in their study sustained a patellar tendon rupture 6 weeks after the procedure [94].

As previously noted, a double-blind RCT by Dragoo et al. [66] comparing PNT (10 needle fenestrations) to PNT + LR-PRP found significant improvement in VISA score only in the PNT + LR-PRP group at 12 weeks. However, both groups reported significant improvement after 26 weeks with no difference found between the two groups. It is important to note, however, that 3 athletes in the PNT group were excluded from the 26-week analysis as they crossed over to the PNT + PRP group after failing treatment with PNT [66].

A similar RCT by Kirshner et al. [93] compared PNT to PNT + LR-PRP in a convenience sample of 40 participants with multiple different tendinopathies. They reported decreasing average pain scores over time for both groups, with lower scores in the PNT group at 6 weeks though no differences between groups noted at 54 or 104 weeks. It's important to note that only 2 participants in this study were diagnosed with PT, and both were included in the PNT group [93].

Though further prospective and comparative studies are warranted to assess the efficacy of PNT for PT, limited evidence suggests that it may improve pain and facilitate return to sport over the intermediate- to long-term. The optimal number of fenestrations, optimal needle gauge, and potential additional benefit of concurrent PRP injection remain uncertain. While heterogeneity exists in post-procedural activity restrictions, in general a progressive rehabilitation protocol should follow this procedure. Due to the associated time to pain improvement, disruption of the patellar tendon, and recovery time involved, this treatment is likely best reserved for the off-season athlete.

Percutaneous Ultrasonic Tenotomy

Percutaneous ultrasonic tenotomy (PUT) is a minimally invasive procedure that involves debridement of tendinopathic tissue utilizing ultrasonic energy and sterile saline irrigation.

Currently, there are no published RCTs evaluating the efficacy and long-term safety of PUT for PT. However, several case reports and case series have reported encouraging outcomes following the use of PUT for the treatment of chronic, refractory PT [95–98].

Nanos and Malanga describe a collegiate basketball player who was able to play 5 months after PUT and PRP injection after previously failing extensive conservative treatment and arthroscopic debridement [95]. Kruse et al. described a collegiate baseball player with bilateral PT who reported complete resolution of symptoms and return to sport 12 weeks after concurrent bilateral PUT [96]. Omodani et al. [97] report a case series of five high-level athletes with eight affected knees treated with PUT. Four of the five athletes (six out of eight affected knees) fully returned to sport at an average time of 3 months and 19 days [97].

The safety of PUT has been evaluated in a cadaveric study, which found that the procedure effectively debrided the intended treatment area without associated damage to the surrounding tissue [99].

Prospective comparative studies and RCTs are needed to further assess safety and long-term efficacy relative to other interventions. However, PUT can be considered in cases of refractory PT. As PUT involves disruption and debridement of the tendinopathic region, several months are typically required for recovery and rehabilitation. This procedure is likely best performed during the off-season or for an athlete already sidelined by PT symptoms.

Operative Treatment

Surgical management is typically reserved for refractory cases, often in elite athletes, who have not responded to 6 to 12 months of conservative treatment [100, 101]. Surgical consideration is particularly relevant in cases where advanced imaging shows partial tendon rupture, calcification, enthesophyte formation, involvement of Hoffa's fat pad, or in patients who have failed exhaustive nonoperative management [52].

While multiple surgical techniques have been described for the treatment of chronic PT, most fall within the categories of either open or arthroscopic treatments [101]. Arthroscopic techniques utilize minimally invasive approaches to excise neovascularized and degenerative tendon tissue. In severe cases, concomitantly performed decortication or resection of the inferior patellar pole may also be considered at the time of arthroscopy [102]. Favorable clinical outcomes have been described across multiple studies evaluating results of arthroscopic treatment of PT. In a retrospective series by Lee et al., arthroscopic patellar tendon debridement and decortication of the inferior pole of the patella produced favorable clinical outcomes in a cohort of 37 athletes with significant improvements in International Knee Documentation Committee (IKDC) score, VISA-P, and Kujala score at a mean of 51 months postoperative [102].

Alfredson et al. reported successful results using an US- and Doppler-guided wide awake local anesthetic no tourniquet (WALANT) arthroscopic approach, with significant improvements in VISA-P scores, 83% surgical satisfaction, and no postoperative complications [100]. Positive outcomes have also been demonstrated after various open surgical treatments, such as tenotomy, inferior patellar pole resection, and suture repair with biologic PRP augmentation [103]. In a retrospective series of 12 patients that underwent open patellar tendon tenotomy and debridement with suture-bridging double-row repair, Zhang et al. demonstrated significantly improved VAS pain and Lysholm scores at minimum 14 months with no reported complications [104]. Similarly, a systematic review by Brockmeyer et al. found that open surgery had an overall success rate of 87% in patients with chronic patellar tendinosis [105]. Notably, arthroscopic treatment was also associated with a very favorable success rate of 91% in their review [105].

Both arthroscopic and open surgical treatments have produced high rates of return to sport in the available literature. A systematic review of 40 studies by Cognetti et al. found an overall return-to-sport rate of 89.8% (96.2% for arthroscopic and 85.8% for open procedures) after surgical treatment for recalcitrant PT [106]. In their analysis, 76.1% of those athletes returned to the same level of competition [106]. Similarly, Sandoval et al. reported a 92.1% overall return-to-sport rate after 3.9 postoperative months in a recent systematic review of 11 studies that included 454 athletes [107]. Another review by Stuhlman et al. found that open and arthroscopic techniques produced return-to-sport rates of 81% and 91%, respectively, with return to play after arthroscopic surgery occurring slightly faster at mean 5 months versus 5.6 months [108]. While some evidence suggests that arthroscopic surgery may provide quicker return to sport compared to open procedures, clinical superiority of either approach has not yet been established, with both techniques producing favorable results overall.

Postoperative complications after surgical treatment for chronic PT are rare, but can include tendon rupture, superficial wound infections, knee stiffness, and sensory complaints or paresthesia [101, 107]. Furthermore, post-surgical wound complications over the patellar tendon may necessitate specialized closure techniques, including plastic surgery. In a systematic review of 1,238 knees that underwent surgery for PT, 36 patients (2.9%) underwent revision surgery, with the most common indication being refractory symptoms [107]. Less commonly performed revision surgeries included arthroscopic lysis of adhesions and wound debridement. While complication rates for surgical management of chronic PT are generally low, surgical intervention carries inherent risks and requires substantial postoperative rehabilitation, all of which should be discussed in detail during perioperative patient counseling [100, 101]. Multiple reviews of available high-level clinical trials have failed to demonstrate superiority of surgical treatment at mid- or long-term follow up compared to placebos or nonoperative measures, such as eccentric exercises and other therapeutic modalities [109, 110]. Notable variability in surgical techniques and postoperative protocols exists among available evidence, further limiting comparison across studies [111].

High-quality evidence comparing surgical outcomes versus current nonoperative treatment protocols remains sparse, creating challenges for establishing ideal surgical indications and treatment consensus [112]. These treatment considerations and clinical outcomes should all be discussed in detail with athletes to help them make informed decisions and create individualized treatment plans. Further RCTs are needed to establish best clinical practices and clarify the role of surgical intervention in the treatment algorithm of chronic PT in athletes.

Limitations

This review has several limitations. Though a comprehensive methodology was used to evaluate the literature, the narrative review format inherently contributes to subjectivity in article selection. Many studies describe a small sample size or a single case, and the variable quality of evidence between different treatment modalities makes it challenging to directly compare interventions. Though most of the available literature reports on the treatment of athletes, some studies included a small number of non-athletes in their sample. Furthermore, many included studies do not delineate between treatment of in-season versus off-season athletes. However, this distinction was outlined in the text in an effort to help guide clinicians with their clinical decision-making as this is an important variable in deciding treatment. While cost was mentioned multiple times in this review, very few if any studies mentioned this as part of the consideration for any given treatment. Nevertheless, cost is an important variable that may impact an athlete's decision on treatment. Several different surgical techniques have been described, though there is no clear consensus on the optimal surgical treatment method. Finally, adverse events were not reported in all studies.

Conclusions

Managing PT in athletes, particularly those with refractory symptoms, remains a clinical challenge. A variety of factors should be considered when counseling athletes and selecting treatment options, including the invasiveness, cost, associated recovery, return-to-sport timeline, and efficacy (Table 2).

Table 2 Factors to consider when comparing treatment options for athletes with PT

	Potential Benefits	Recovery	Risks/Common Side Effects	Accessibility	Cost	In-Season	Other
Physical Therapy	Significant long-term pain relief in many cases, high RTS rates in many cases	-	Exacerbation of symptoms	Available to most athletes	Low, typically covered by insurance	Yes	Progressive tendon loading and eccentrics are preferred modalities
Bracing and Taping	Modest short-term pain relief, uncertain impact on RTS	-	Taping may modestly reduce maximum jump height	Available to most athletes	Low, out-of-pocket cost $10–30	Yes	
Topical GTN	Modest short-term pain relief, uncertain impact on RTS	-	Headache, skin rash, lightheadedness	Available to most athletes	Low, insurance coverage varies (out-of-pocket cost $1–2 per patch)	Yes	
ESWT	Moderate immediate pain relief, mixed outcomes for long-term pain relief, mixed RTS outcomes	Can return to sport using a pain monitoring model	Mild pain/discomfort during treatment, redness, bruising, paresthesia, skin irritation	May be limited locally for some athletes	Out-of-pocket cost $100–400 per session	Yes	Treatment typically consists of 3–5 sessions. Variable protocols.
Platelet Rich Plasma	Mixed outcomes for long-term pain relief and RTS	≥ 6 weeks (≥ 12 weeks if tenotomy performed)	Pain at injection site up to several weeks, swelling	May be limited locally for some athletes	Most insurance companies do not cover PRP. Out-of-pocket cost may range from $500–2500	No	
Hyaluronic Acid	Limited evidence of moderate pain relief and mixed RTS outcomes	Uncertain	Pain at injection site	Limited for most athletes due to off-label use	Limited insurance coverage. Out-of-pocket cost may range from $500–2500 per injection	Due to limited data, potential high out-of-pocket cost, and onset of action, may be better suited for off season use. In-season may be considered if peritendinous.	Studies report up to 10 repeat injections
High-Volume Injection	Mixed outcomes for long-term pain relief and RTS	≥ 5 days	Pain at injection site	May be limited locally for some athletes due to provider experience	May be covered by some insurance providers. Out-of-pocket cost may range from $500–2500	Yes	Various options for injection
PrT	Limited evidence of moderate intermediate-term pain relief, uncertain RTS outcomes	1–2 days if peritendinous, longer if intratendinous	Pain at injection site, swelling	May be limited locally for some athletes	Limited insurance coverage. Out-of-pocket cost may range from $150–500 per injection	Potentially if peritendinous; No if intratendinous	Multiple injections typically required
BMAC	Limited evidence of moderate long-term pain relief, uncertain RTS outcomes	Uncertain	Pain at injection site, hematoma formation at aspiration site, swelling, infection, bleeding	Limited due to off-label use	Out-of-pocket cost may range from $1000-4500	No	
Tendon Scraping	Limited evidence of short-term pain relief and high RTS rates	≥ 3 days	Pain at procedure site, accidental tenotomy	May be limited locally for some athletes	May not be covered by insurance	Yes	

Table 2 (continued)

	Potential Benefits	Recovery	Risks/Common Side Effects	Accessibility	Cost	In-Season	Other
PNT	Limited evidence of long-term pain relief and high RTS rates.	≥ 12 weeks	Pain at procedure site, accidental tendon injury	May be limited locally for some athletes	May not be covered by insurance	No	
PUT	Limited evidence of moderate long-term pain relief and high RTS rates	≥ 12 weeks	Pain at procedure site, accidental tendon injury	May be limited locally for some athletes	Maynot be covered by insurance	No	
Surgery	Significant long-term pain relief, high RTS rates	≥ 5 months	Pain at procedure site, infection, knee stiffness, paresthesia, revision surgery	Available to most athletes	Covered by insurance	No	

Given the complexity of PT management in athletes, a conceptual framework has been provided to aid clinicians in their clinical decision making (Fig. 1). While this framework is not exhaustive and cannot account for every clinical phenotype of PT, it is intended to complement the accompanying literature review and serve as a practical guide for clinicians. The variability of this condition, timeline of competitive season, and multiple treatment options make

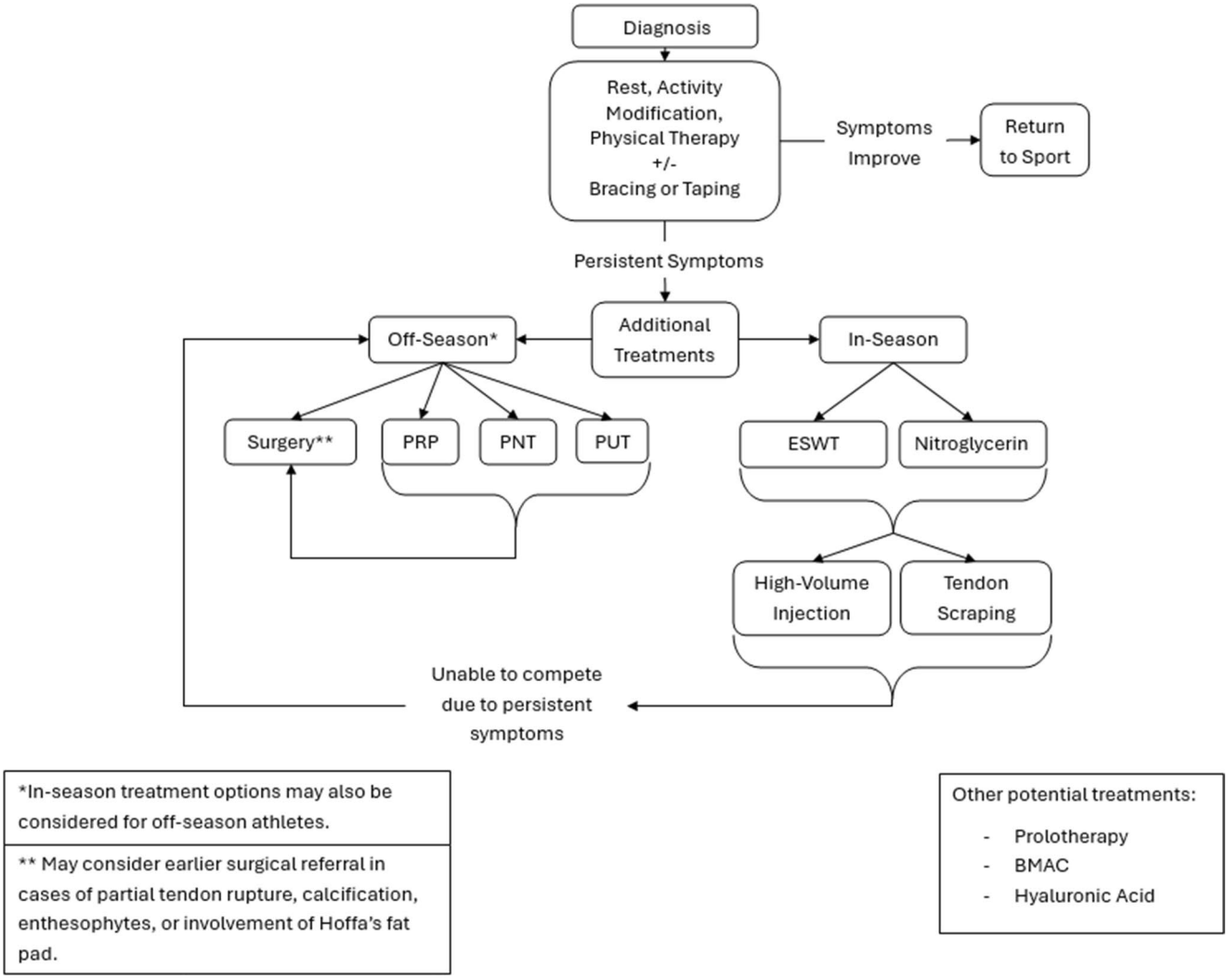

Fig. 1 Framework for the management of in-season and off-season athletes with PT

it difficult to establish a standardized treatment protocol. However, this approach combined with the clinician's judgment may be utilized to create an individualized treatment plan for an athlete.

In most cases, management should begin with a structured course of physical therapy. Bracing and taping can be trialed as adjuncts based on athlete preference and comfort. FSW, RPW, and nitroglycerin patches are reasonable non-invasive options for those not adequately improving with initial rehabilitation. Tendon scraping and HVI are safe options for in-season athletes as both procedures maintain the integrity of the patellar tendon. In contrast, PUT, PNT, and intratendinous PRP injection should be reserved for off-season athletes due to their recovery periods. Surgical referral should be considered for athletes with symptoms refractory to 6 to 12 months of conservative treatment.

In summary, multiple non-operative and operative treatment options exist for managing PT in athletes, though comparative data on efficacy is limited. The treatment plan should be individualized based on the athlete's goals and preferences and informed by a nuanced understanding of the literature.

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    - This long-term follow-up study found that nearly one quarter of athletes with PT did not report adequate improvement with therapy alone, demonstrating the need for additional treatment options for refractory cases.
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    - This systematic review of 8 comparative studies found statistically significant benefit in favor of PRP compared to ESWT but not compared to non-PRP injections, highlighting the need for further high-quality RCTs and comparative studies to clarify the efficacy of PRP and non-PRP injections for the treatment of PT.
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    - This systematic review of 40 studies reported very high return-to-sport rates after arthroscopic and open surgical treatment for recalcitrant PT.
- Rhim HC, Singh M, Maffulli N, Saxena A, Leal C, Gerdesmeyer L, et al. Recommendations for use of extracorporeal shockwave therapy in sports medicine: an international modified Delphi study. Br J Sports Med. 2025. doi: https://doi.org/10.1136/bjsports-2024-109082.
    - This study presents updated expert recommendations on ESWT terminology, indications, procedural aspects, postprocedural considerations, contraindications, and side effects.

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